

ST HELENA, Saint Helena

WMO: 619010

Lat: 15.9422S

Lon: 5.6676W

Elev: 436

StdP: 96.20

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	13.6	14.0	9.3	7.6	16.4	9.9	8.0	16.4	14.1	15.6	13.1	15.4	6.5	130	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	3.7	23.8	19.4	23.2	19.3	22.7	19.1	21.0	21.9	20.6	21.7	20.2	21.4	5.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.6	16.1	21.3	20.2	15.7	21.0	19.9	15.4	20.8	62.6	22.0	61.3	21.7	60.2	21.4	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.8	10.1	12.3	25.8	0.8	1.6	11.7	27.0	11.3	28.0	10.8	28.9	10.2	30.1		
(5)			10.6	22.1	0.8	1.2	10.0	22.9	9.5	23.6	9.1	24.3	8.5	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	19.9	21.0	21.3	20.6	19.4	17.8	16.7	16.0	16.4	17.1	18.4	(6)	
(7)		DBStd	2.07	0.84	0.77	0.75	0.78	0.92	0.83	0.85	0.84	0.74	0.80	0.84	0.94	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	323	1	0	0	0	2	20	50	72	70	62	37	11	(9)
(10)		CDD10.0	3056	307	308	349	317	293	233	209	187	180	197	214	261	(10)
(11)		CDD18.3	337	49	75	91	67	36	4	0	0	0	0	1	13	(11)
(12)		CDH23.3	46	3	17	20	4	1	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	1	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	6.5	6.1	5.8	6.1	6.2	5.7	6.2	6.6	7.3	7.3	7.4	6.9	6.4	(14)
(15)	Precipitation	PrecAvg	498	36	41	52	43	41	60	56	39	26	19	28	(15)	
(16)		PrecMax	733	84	120	104	82	115	132	124	101	82	60	41	110	(16)
(17)		PrecMin	339	7	10	18	9	11	2	4	16	4	5	3	8	(17)
(18)		PrecStd	97	20	20	23	22	23	28	27	21	17	14	10	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	24.7	24.7	24.0	22.9	21.2	20.2	19.2	19.3	20.1	20.9	22.1	(19)
(20)			MCWB	18.6	19.4	20.0	19.3	18.3	16.9	15.7	15.2	15.4	16.0	16.5	17.1	(20)
(21)		2%	DB	22.7	23.7	23.9	23.1	22.1	20.4	19.2	18.5	18.4	18.9	19.9	21.2	(21)
(22)			MCWB	18.3	19.3	19.8	19.1	18.1	16.5	15.4	14.9	15.0	15.5	16.1	17.1	(22)
(23)		5%	DB	22.1	23.2	23.2	22.5	21.5	19.7	18.7	17.8	17.7	18.2	19.2	20.6	(23)
(24)			MCWB	18.2	19.3	19.6	19.0	17.9	16.3	15.3	14.7	14.8	15.3	15.8	17.0	(24)
(25)		10%	DB	21.5	22.6	22.7	22.0	20.9	19.2	18.1	17.3	17.2	17.6	18.5	20.0	(25)
(26)			MCWB	18.1	19.2	19.5	18.9	17.7	16.1	15.1	14.6	14.7	15.2	15.7	16.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.1	21.2	21.7	21.1	19.7	18.3	17.0	16.2	16.5	17.0	17.5	19.0	(27)
(28)			MCDWB	21.5	22.2	22.5	21.8	21.1	19.5	18.3	17.4	17.7	18.5	19.0	20.0	(28)
(29)		2%	WB	19.6	20.7	21.1	20.6	19.2	17.6	16.5	15.8	15.9	16.4	16.9	18.3	(29)
(30)			MCDWB	20.9	21.8	22.0	21.5	20.6	19.0	17.8	16.9	16.9	17.7	18.4	19.5	(30)
(31)		5%	WB	19.2	20.4	20.7	20.2	18.8	17.2	16.1	15.5	15.6	16.0	16.5	17.9	(31)
(32)			MCDWB	20.5	21.6	21.8	21.2	20.3	18.6	17.5	16.7	16.6	17.2	18.0	19.2	(32)
(33)		10%	WB	18.9	20.1	20.4	19.8	18.4	16.8	15.8	15.2	15.2	15.6	16.2	17.6	(33)
(34)			MCDWB	20.2	21.3	21.6	21.0	20.0	18.3	17.3	16.5	16.4	16.9	17.6	19.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.0	3.9	3.7	3.5	3.5	3.3	3.2	2.9	3.0	3.2	3.5	3.8	(35)
(36)			MCDBR	4.5	4.5	4.2	3.9	4.1	3.9	3.7	3.4	3.4	3.7	4.0	4.2	(36)
(37)			MCWBR	2.1	2.1	2.0	2.2	2.4	2.5	2.5	2.3	2.3	2.1	2.2	2.3	(37)
(38)		5% WB	MCDBR	4.0	3.8	3.5	3.3	3.7	3.4	3.3	3.0	3.1	3.3	3.6	3.8	(38)
(39)			MCWBR	2.1	2.0	2.0	2.1	2.4	2.5	2.5	2.4	2.3	2.1	2.2	2.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.344	0.342	0.345	0.329	0.316	0.311	0.330	0.355	0.407	0.379	0.335	0.335	(40)	
(41)		taud	2.585	2.616	2.618	2.657	2.665	2.650	2.574	2.483	2.298	2.415	2.586	2.601	(41)	
(42)		Ebn at Noon	999	992	968	950	930	918	905	910	894	945	1001	1008	(42)	
(43)		Edh at Noon	106	102	99	89	83	81	90	105	134	123	105	104	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.93	6.99	6.45	5.52	4.76	4.14	4.20	4.40	4.71	5.03	5.74	6.41	(44)	
(45)		RadStd	0.43	0.23	0.31	0.33	0.21	0.20	0.33	0.32	0.30	0.35	0.41	0.39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	(e) +0.20	(f) +0.38	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page